



# भारतीय मानक ब्यूरो BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

## व्यापक परिचालन मसौदा

हमारा संदर्भ — सीईडी 09/टी – 55

24 अगस्त 2026

तकनीकी समिति — टिम्बर एवं टिम्बर स्टोर्स विषय समिति, सीईडी 09

प्राप्तकर्ता:

- 1 सिविल इंजीनियरिंग विभाग परिषद, सीईडीसी के सभी सदस्य
- 2 टिम्बर एवं टिम्बर स्टोर्स विषय समिति, सीईडी 09 के सभी सदस्य
- 3 सीईडी 09 के अंतर्गत पैनल एवं वर्किंग ग्रुप के सभी सदस्य
- 4 रूचि रखने वाले अन्य निकाय।

प्रिय महोदय/महोदया,

कृपया निम्नलिखित मसौदा संलग्न देखें:

| प्रलेख संख्या       | शीर्षक                                                                                                                                             |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| सीईडी 09 (35184) WC | पेड़ काटने वाली कुल्हाड़ियों और हाथ वाली कुल्हाड़ियों के लिए लकड़ी के हैंडल — विशिष्टि का भारतीय मानक मसौदा<br>(द्वसरा पुनरीक्षण)<br>(ICS: 79.040) |

कृपया इस मसौदे का अवलोकन करें और अपनी समितियाँ यह बताते हुए भेजे कि यह मसौदा प्रकाशित हो तो इस पर अमल करने में, आपको व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं।

**समितियाँ भेजने की अंतिम तिथि — 08 अक्टूबर 2026**

सम्मति यदि कोई हो तो कृपया अधोहस्ताक्षरी को उपरिलिखित पते पर संलग्न फॉर्मेट में भेजें या [ced9@bis.gov.in](mailto:ced9@bis.gov.in); [nishikant.singh@bis.gov.in](mailto:nishikant.singh@bis.gov.in) पर ईमेल कर दें या समितियाँ बीआईएस ई-गवर्नेंस पोर्टल, [www.manakonline.in](http://www.manakonline.in) के माध्यम से ऑनलाइन भी भेजी जा सकती हैं।

यदि कोई सम्मति प्राप्त नहीं होती है अथवा सम्मति में केवल भाषा संबंधी त्रुटि हुई तो उपरोक्त प्रलेख को यथावत अंतिम रूप दे दिया जाएगा। यदि सम्मति तकनीकी प्रकृति की हुई तो विषय समिति के अध्यक्ष के परामर्श से अथवा उनकी इच्छा पर आगे की कार्यवाही के लिए विषय समिति को भेजे जाने के बाद प्रलेख को अंतिम रूप दे दिया जाएगा।

यह प्रलेख भारतीय मानक ब्यूरो की वेबसाइट [www.bis.gov.in](http://www.bis.gov.in) पर भी उपलब्ध हैं।

धन्यवाद।

|                                                                 |
|-----------------------------------------------------------------|
| भवदीय                                                           |
| ह-/                                                             |
| द्विपायन भद्र                                                   |
| वै. 'एफ' / वरिष्ठ निदेशक और प्रमुख<br>(सिविल इंजीनियरिंग विभाग) |

संलग्न : उपरिलिखित



भारतीय मानक ब्यूरो  
BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

**WIDE CIRCULATION DRAFT**

Reference — CED 09/T-55

Dated — 24 August 2026

TECHNICAL COMMITTEE — TIMBER AND TIMBER STORES SECTIONAL COMMITTEE, CED 09

**ADDRESSED TO:**

- 1 All Members of Civil Engineering Division Council, CEDC
- 2 All Members of Timber and Timber Stores Sectional Committee, CED 09
- 3 All Members of Panels and Working Groups under CED 09
- 4 All other interested

Dear Madam/Sir,

| Doc. No.          | Title                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| CED 09 (35184) WC | Draft Indian Standard Wooden Handles for Felling Axes and Hand Axes —<br>Specification<br>(Second Revision)<br>(ICS: 79.040) |

Kindly examine the draft and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

**Last Date for comments — 08 October 2026**

Comments if any, may please be made in the attached format and mailed to the undersigned at the above address or preferably through e-mail to [ced9@bis.gov.in](mailto:ced9@bis.gov.in); [nishikant.singh@bis.gov.in](mailto:nishikant.singh@bis.gov.in). The comments may preferably be shared in the prescribed template through the Manak Online portal at [www.manakonline.in](http://www.manakonline.in). Alternatively, the comments may be sent through the attached format for consideration by the BIS' Sectional Committee for necessary action.

In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case comments, technical in nature are received, then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

The document is also hosted on BIS website [www.bis.gov.in](http://www.bis.gov.in).

**Thanking You**

|                                                                              |
|------------------------------------------------------------------------------|
| <b>YOURS FAITHFULLY</b>                                                      |
| Sd/-                                                                         |
| <b>Dwaipayan Bhadra</b>                                                      |
| <b>Sc. 'F' / Senior Director and Head<br/>(Civil Engineering Department)</b> |

**ENCLOSED— As Above**

## FORMAT FOR SENDING COMMENTS ON BIS DOCUMENTS

(Please use A4 size sheet of paper only and type within fields indicated. Comments on each clause/subclause/table/fig etc. be started on a fresh box. Information in column 5 should include reasons for the comments, and those in column 4 should include suggestions for modified wording of the clauses when the existing text is found not acceptable. Adherence to this format facilitates Secretariat's work) {Please e-mail your comments to [ced9@bis.gov.in](mailto:ced9@bis.gov.in), [nishikant.singh@bis.gov.in](mailto:nishikant.singh@bis.gov.in)

|                                       |                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Doc. No.                              | Doc: CED 09 (35184) WC                                                                                                          |
| Title                                 | Draft Indian Standard Wooden Handles for Felling Axes and Hand Axes — Specification ( <i>Second Revision</i> )<br>(ICS: 79.040) |
| Last Date for Comments                | 08 October 2026                                                                                                                 |
| Name of the Commentator /Organization |                                                                                                                                 |

[illegible]

**BUREAU OF INDIAN STANDARDS**

**DRAFT FOR COMMENTS ONLY**

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*DRAFT INDIAN STANDARD*

**WOODEN HANDLES FOR FELLING AXES AND HAND AXES —  
SPECIFICATION**

*(Second Revision)*

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Timber and Timber Stores  
Sectional Committee, CED 09

Last Date for Comments:  
**08 October 2026**

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**FOREWORD**

*(Formal Clause to be added later)*

The general requirements for wooden tool handles are covered by IS 620. A series of standards have been prepared for various hand tools and for use with these hand tools a series of standards for the corresponding wooden handles are being prepared. This standard deals with the wooden handles for felling axes and hand axes conforming to IS 703.

This Indian Standard was first published in 1964 and subsequently revised in 1975. This standard has been taken up for revision with a view to bring its provisions in line with current practices. In this revision, the following modifications have been incorporated:

- a) BIS certification marking clause has been updated;
- b) References to various Indian Standards have been updated; and
- c) The standard has been aligned with the latest style and format of Indian Standards, with editorial and structural changes to improve clarity and ease of use.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded-off value should be the same as that of the specified value in this standard.

**BUREAU OF INDIAN STANDARDS****DRAFT FOR COMMENTS ONLY**

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*DRAFT INDIAN STANDARD***WOODEN HANDLES FOR FELLING AXES AND HAND AXES —  
SPECIFICATION**

*(Second Revision)*

Timber and Timber Stores  
Sectional Committee, CED 09

Last Date for Comments:  
**08 October 2026**

**1 SCOPE**

This standard covers the requirements of wooden handles for the following four types of axes conforming to IS 703:

- a) Axe, hand;
- b) Axe, felling;
- c) Axe, felling, estate pattern; and
- d) Axe, felling, trade pattern.

**2 REFERENCES**

The standards given below contains provisions, which through reference in this text, constitute provisions of this standard. At the time of publication the editions indicated are valid. All standards are subject to revision, and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards:

| <i>IS No.</i> | <i>Title</i>                                                                           |
|---------------|----------------------------------------------------------------------------------------|
| IS 620 : 1985 | Specification for Wooden Tool Handles General Requirements<br><i>(Fourth Revision)</i> |
| IS 703 : 1999 | Axes — Specification <i>(Second Revision)</i>                                          |

**3 TERMINOLOGY**

For the purpose of this standard, the definitions given in IS 620 shall apply.

**4 SPECIES OF TIMBER**

**4.1 Handles for Hand Axes** — These shall be manufactured from any of the timber species specified for Class III handles in IS 620.

**4.2 Handles for Felling Axes** — These shall be manufactured from any of the timber species specified for Class I handles in IS 620.

**4.3 Handles for Felling Axes, Estate and Trade Pattern** — These shall be manufactured from any of the timber species specified for Class I handles in IS 620.

## 5 REQUIREMENTS OF BLANKS

**5.1** The blanks shall be well-seasoned to a moisture content not exceeding 15 percent and not less than 8 percent. The blanks shall be free from centre heart (pith), any kind of decay (rot), checks, shakes, splits (except hair splits), borer holes, warp and any other defect except those permitted in the, finished handles in IS 620.

**5.2** The green blanks, if so desired, may also be given treatment as given in Annex A to prevent cracking and splitting.

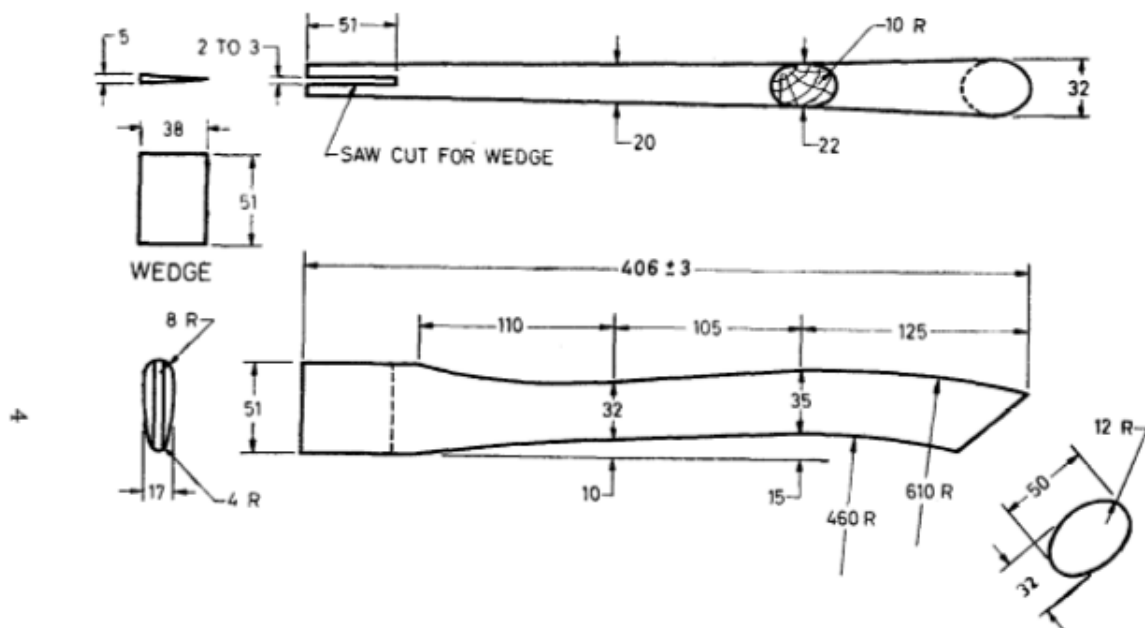
## 6 MANUFACTURE

**6.1** The handles shall be manufactured from selected blanks prepared as above and shaped as shown in Fig. 1 to 3.

**6.2** It is recommended that immediately after manufacture, all the handles be dipped in raw linseed oil at room temperature as this treatment retards changes in the moisture content.

## 7 DIMENSIONS AND TOLERANCES

**7.1 Dimensions** — The handles shall conform to the dimensions as shown in Fig. 1 to 3.



NOTE — Wedge shall be dipped in glue and hammered tight.

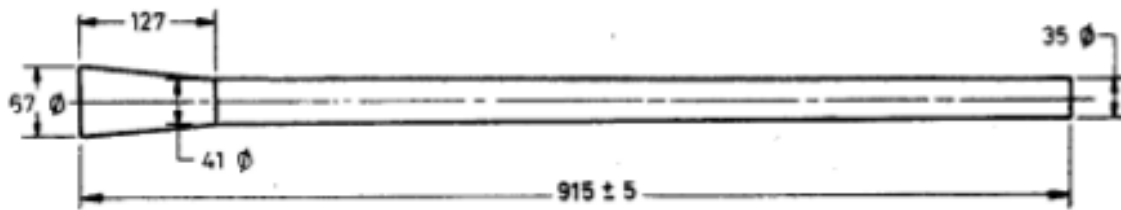
All dimensions in millimetres.

Fig. 1 Wooden Handle for Axes, Hand



All dimensions in millimetres.

Fig. 2 Wooden Handle for Axes, Felling, Trade Pattern



All dimensions in millimetres.

Fig. 3 Wooden Handle for Axes, Felling, Estate Pattern

**7.2 Tolerances** — The following tolerances shall be permissible on the diameter of the handles:

- |                                     |                  |
|-------------------------------------|------------------|
| a) Where the head of the tools fits | + 3 mm<br>– 0 mm |
| b) At other places                  | ± 2 mm           |

## 8 OTHER REQUIREMENTS

**8.1 Workmanship and finish, permissible defects, treatment, and sampling** shall be as specified in IS 620.

## 9 TESTS

**9.1** In addition to the tests specified in IS 620, the following tests shall also be carried out on the finished handles selected in accordance with the sampling procedure given in IS 620.

**9.2 Drop Test** — The handle shall be held horizontally and dropped from a height of 1.25 m on a concrete surface, 10 times continuously.

**9.3 Impact Test** — The handle shall be fitted with a dummy head of approximately the same weight as that of the tool. It shall then be struck with a reasonable force, on a hardwood surface, 10 times continuously.

**9.4** At the end of the tests specified in 9.2 and 9.3, there shall be no visible damage, chips or cracks at any portion of the handle.

## 10 SAMPLING

**10.1** The method of drawing representative samples of handles and criteria for conformity shall be as prescribed in IS 620.

**11 MARKING**

**11.1** Unless otherwise specified, each handle shall be legibly and indelibly marked with the manufacturer's name, initials or recognized trade-mark, if any, name of timber species and the year of manufacture. In addition, the type of the axe for which the handle is required, shall be marked as shown below:

|    | <i>Type</i>                  | <i>Marking Code</i> |
|----|------------------------------|---------------------|
| a) | Axe, hand                    | AH                  |
| b) | Axe, felling                 | AF                  |
| c) | Axe, felling, estate pattern | AFEP                |
| c) | Axe, felling, trade pattern  | AFTP                |

**11.1.1 BIS Certification Marking**

The product(s) may be marked with Standard Mark as per the conformity assessment schemes governed by the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations made there under. The details of conditions for the license may be obtained from the Bureau of Indian Standards.



## **ANNEX A**

*(Clause 5.2)*

### **TREATMENT OF GREEN BLANKS**

**A-1** For the treatment, the green blanks shall be converted to the sizes as close as possible to the shape and dimensions of the handles (allowing some oversize for shrinkage and final finishing) and immediately immersed in a 40 percent (w/w) solution of polyethylene glycol (PEG)-1500 or 60 percent (w/w) solution of urea for approximately 24 hours.

**A-2** The blanks after the treatment as mentioned above shall be air-seasoned or kiln-seasoned in the usual manner until the desired moisture content is attained. The finishing of handles shall be done at the end.